

ANIRUDH SURABI

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TECHNICAL SKILLS

Languages	Python, JavaScript, SQL
ML / Deep Learning	TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, CNNs, Transfer Learning, ResNet50, DenseNet201, VGG16, AlexNet, U-Net, Pix2Pix, HDBSCAN, TF-IDF
Parallel Computing	OpenMP, CUDA, GPU-Accelerated Computing, HPC
Web & Databases	React, HTML, CSS, Framer Motion, Material UI, PostgreSQL, SQLAlchemy, ORM
Tools & Platforms	Git, GitHub, Linux/Unix, Jupyter Notebook, VS Code

EDUCATION

M.S. Computer Science — Loyola University Chicago • **GPA: 4.0** May 2026
B.S. Information Technology — GRIET (Gokaraju Rangaraju Institute of Engineering and Technology) August 2023

EXPERIENCE

Graduate Research Assistant | Loyola University Chicago Chicago, IL • Mar 2025 – Aug 2025

- Designed and trained CNN classification models (AlexNet, ResNet50, DenseNet201, VGG16) for physician-rated synthetic medical image quality scoring; fine-tuned with Dropout and L2 regularization to reduce overfitting.
- Built a Pix2Pix image-to-image translation pipeline generating high-fidelity synthetic medical images from clinical datasets, expanding training data for downstream models and advancing the team toward peer-reviewed publication.

Machine Learning Intern | Rinex Technology Remote • Mar 2022 – Apr 2022

- Completed hands-on ML training covering supervised/unsupervised learning, model evaluation, and predictive modeling workflows; earned Rinex Technology ML Certification.

PROJECTS

Parallelizing Conway's Game of Life — Python, NumPy, OpenMP, CUDA, HPC

- Achieved 13.57× CPU speedup (8 threads) and 139× GPU speedup over serial baseline on NVIDIA RTX 5070 Ti across 268M-cell grids; analyzed memory bandwidth bottlenecks and superlinear cache-effect scaling.

Brain Tumor Detection — Deep Learning — TensorFlow, Keras, ResNet50, DenseNet201, VGG16

- Built transfer learning pipelines for binary and multi-class MRI classification, exceeding 85% test accuracy through systematic data augmentation and hyperparameter tuning.

Real-Time Pothole Detection System — Python, TensorFlow/Keras, CNNs, OpenCV

- Engineered a CNN achieving 95% accuracy, sub-100ms latency, and 3% false positive rate; integrated into vehicle navigation systems and validated through field testing.

Scene Understanding in Construction Sites — Python, TensorFlow/Keras, CNNs, Computer Vision

- Trained object detection model reaching 97% accuracy and 90% average precision on live construction site deployments; built worker-facing UI reducing material inventory errors.

Job Market Clustering & Ghost Job Detection — Python, Scikit-learn, HDBSCAN, KMeans, TF-IDF

- Applied HDBSCAN and association rule mining to LinkedIn data to detect ghost postings and surface skill demand patterns; built interactive dashboard evaluated via silhouette score and lift metrics.

CERTIFICATIONS

Python — HackerRank **Machine Learning** — Rinex Technology **Python Computing** — NPTEL / IIT **AI Training** — Oracle Academy